# House Price Prediction — Task 2

Feature Engineering, Model Optimization & Performance Comparison

## Objective

Building on the baseline model from Task 1, this task trains and compares multiple regression algorithms on the California Housing dataset. The goal is to apply proper feature scaling, evaluate each model using standard regression metrics, and select the best-performing model with justification.

## Dataset

**California Housing Dataset** (scikit-learn) — 20,640 records. Target variable is median house value. Input features include median income, house age, average rooms, average bedrooms, population, average occupancy, and location (latitude/longitude).

## Methodology

- Loaded the dataset and separated it into features (X) and target (y).
- Applied **StandardScaler** to scale all input features to a common range.
- Split the data into training and test sets (80% / 20%, random_state = 42).
- Trained three regression models: Linear Regression, Ridge Regression (alpha = 1.0), and a Decision Tree Regressor (max_depth = 5).
- Evaluated each model on the test set using RMSE (Root Mean Squared Error) and $R^2$ Score.
- Compared results in a summary table and selected the best model based on the highest $R^2$ and lowest RMSE.

## Why Three Models?

- **Linear Regression** — serves as the baseline model.
- **Ridge Regression** — adds L2 regularization to help reduce overfitting.
- **Decision Tree Regressor** — captures non-linear relationships that linear models miss.

## Model Performance Comparison

Each model was evaluated on the same held-out test set. Lower RMSE indicates smaller average prediction error; higher $R^2$ indicates the model explains more of the variance in house prices.

| Model | RMSE | $R^2$ Score |
| --- | --- | --- |
| Linear Regression | 0.7456 | 0.5758 |
| Ridge Regression | 0.7456 | 0.5758 |
| **Decision Tree** | **0.7030** | **0.6228** |

The Decision Tree row is highlighted — it achieves the lowest RMSE and highest $R^2$ among the three models.

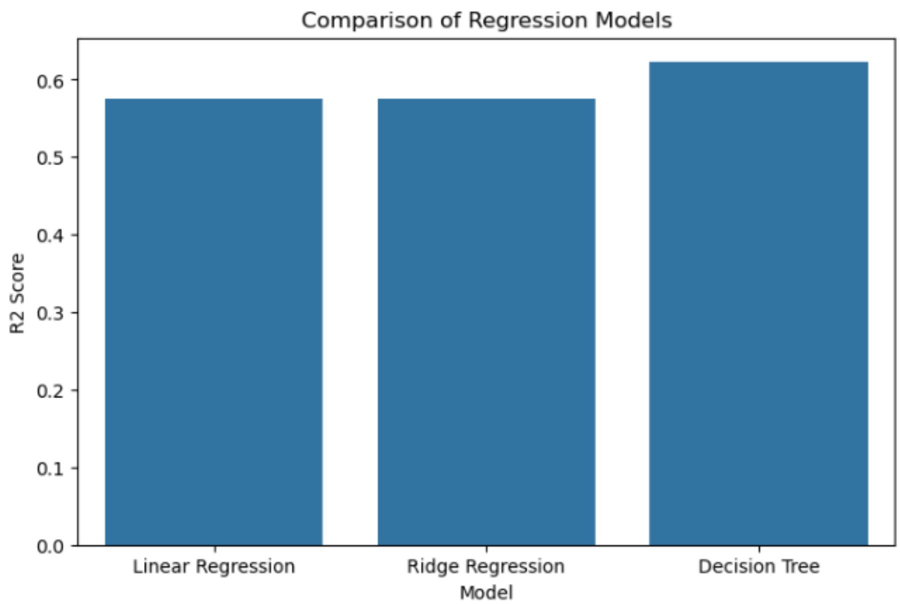


Figure 1: $R^2$ Score comparison across the three regression models.

## Visual Performance Validation

The following plots visualize prediction accuracy and error distribution for the model used in visual validation. Points close to the diagonal in the actual-vs-predicted plot indicate accurate predictions; residuals scattered evenly around zero indicate a well-behaved model.

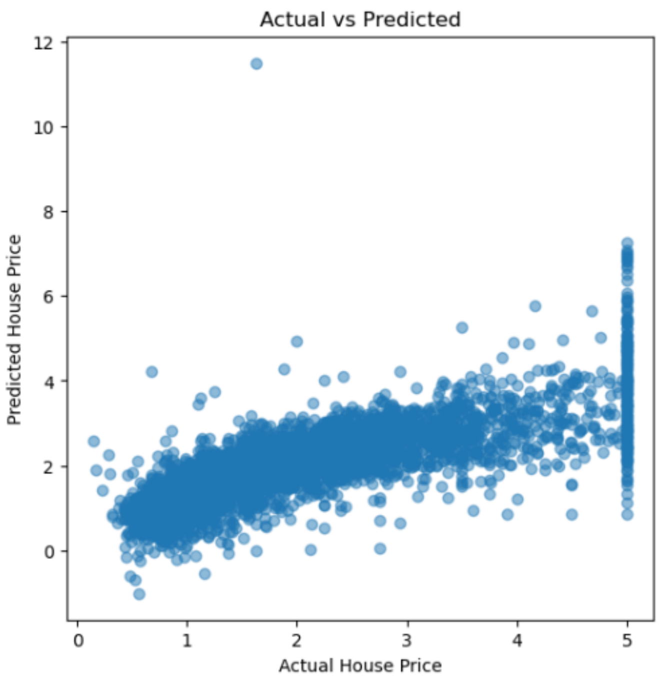


Figure 2: Actual vs. Predicted house prices.

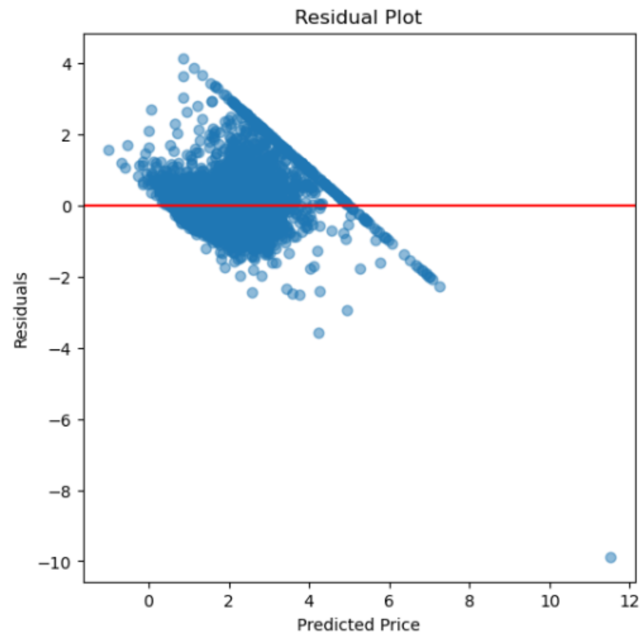


Figure 3: Residual plot (errors vs. predicted price).

Predictions track the diagonal reasonably well at lower price ranges but spread out at higher actual prices — a common pattern where the model underestimates the most expensive homes. Residuals are centered near zero with a wider spread as predicted price increases, and one notable outlier is visible near a predicted price of 11.5.

## Best Performing Model

| | |
|---|---|
| **Model** | Decision Tree |
| **RMSE** | 0.7030 |
| **$R^2$ Score** | 0.6228 |

## Conclusion

- Three regression models — Linear Regression, Ridge Regression, and Decision Tree — were trained and evaluated on the California Housing dataset.
- The comparison was based on RMSE and $R^2$ Score, using an identical, scaled train/test split for all three models.
- Ridge Regression performed almost identically to plain Linear Regression, indicating that regularization had little effect on this dataset.
- The Decision Tree Regressor produced the lowest RMSE (0.7030) and highest $R^2$ (0.6228), making it the best-performing model of the three for predicting California house prices.
- The model with the highest $R^2$ Score and lowest RMSE is considered the best model — by this criterion, the Decision Tree is selected as the final model.

## Key Learning Outcomes

- Applying preprocessing techniques (feature scaling) correctly.
- Comparing multiple Machine Learning models objectively using consistent metrics.
- Interpreting RMSE and $R^2$ Score professionally.
- Justifying a model selection decision using measurable evidence rather than assumption.